A smaller cloud foundation. The remaining features land in stages.

62 commits including this review. Deployment is commit 3; this walkthrough groups the work into ten stages.

01 / FOUNDATION

Preserve main

One serial executor, source checks, private memory and Clockwork schedules.

02 / DEPLOYMENT

Prove the image

Build and boot without writable host storage before optional adapters.

03 / EXTENSIONS

Add independent features

Adapters, queries, recording, local artifacts, embedding and assurance.

Hardest problem: remove features and their consumers together while keeping each retained stage buildable.

Merge review: The branch is ready for staged review; it has not been merged or deployed to production.

Cloud state stays in memory.

The host filesystem supplies startup inputs. It is not a persistence backend.

SESSIONS

Private roots and history

The central actor retains each session’s files and messages between turns. Every command starts with fresh lexical scope.

CONFIGURATION

Actor-held settings

Source edits, secrets and users stay in memory. API updates leave startup files unchanged.

CLUSTER

No disk-first detour

No JSON snapshots, local SQLite store or writable volume. Cluster replication is separate work.

Hardest problem: remove storage paths, save callbacks and restart assumptions from execution, APIs and tests.

Merge review: A fresh owner does not restore local state. Replication, failover and all-owner-loss behavior still need a cluster design.

Check source before it runs.

Extend main’s actor and policy model with trusted workspace configuration, triggers, timers and the shell/API.

TYPES

Source-derived checks

Check root source and imported code before configuration effects. Do not reconstruct types from captured runtime values.

POLICY

Pass / Mock

Explicit effects use pure gates. Ordinary agent replies and final values still reply directly.

SCHEDULING

Main’s Clockwork

Reuse main’s startup scheduler and next-occurrence helper. Configured timers retain policy and destination.

Hardest problem: keep session identity, reply routing and closure handling consistent through one executor.

Merge review: Keywords stay reserved; use on_input. No separate Reject concept, custom cron parser or public cloud helper package.

Deployment is commit 3: 7f639db8.

The image ships read-only startup configuration and main’s policy and native trigger sources.

BUILD

Independent checkpoint

No feeds, operator pages, cloud databases, query prerequisites or configured adapter implementations.

BOOT

Read-only root

Docker starts with --read-only. The root endpoint returns 200 and unauthenticated /api/me returns 401.

VERIFY

110 cloud tests

Strict build passes. Container filesystem changes are empty; mutable state remains in actors.

Hardest problem: remove the entrypoint’s /data initialization and every runtime dependency on writable storage.

Merge review: Provide front-door authentication before public use. The example name-only login still makes the first user an administrator.

Optional integrations follow deployment.

Network secrets are the immediate next feature; example-chat, WhatsApp and native extensions follow.

CREDENTIALS

Independent policy package

Load actor-held secrets in trusted configuration. Env returns placeholders; Fetch substitutes only for allowed HTTPS hosts.

CHAT

Example-chat / WhatsApp

Move implementations, runtime wiring and tests together. Keep the standalone chat application separate.

NATIVE

Extend main’s adapters

Add Slack/Discord thread and reaction hooks while preserving their checked startup triggers.

Hardest problem: avoid leaving early consumers behind when moving adapters, and separate session IDs from reply addresses.

Merge review: Service feeds and generic service webhook integrations are removed. Chat webhook plumbing remains for WhatsApp.

Queries remain a later language feature.

Typed query tables, fixed-point evaluation and SQL planning do not block cloud deployment.

EVALUATION

Indexed query tables

Infer table types and evaluate recursive rules with indexed fixed-point processing.

PLANNING

SQL translation

Plan evaluated tables as joins; preserve the common interfaces for hosts that support SQLite.

HOSTS

Keep local SQLite local

CLI/browser SQLite and executable examples remain. Cloud SQLite and dataclips are removed.

Hardest problem: place broad local dependencies and compiler metadata at their first real consumers.

Merge review: No retained query commit installs a disk-backed database handler in packages/cloud.

Make execution inspectable and reusable.

Retained shells, resumable execution, trajectories and embedding build on the shared interpreter.

EXECUTION

Consistent semantics

Align interpreter/compiler behavior, masks, shallow handlers and retained lexical binding order.

EVIDENCE

Record and replay

Capture effects, inspect trajectories, pin dependencies and redact secrets in portable archives.

HOSTS

Embed the runtime

Expose native host APIs, provider transports and integrations for terminal and application hosts.

Hardest problem: preserve ordering, cancellation and authority while moving execution across runtimes.

Merge review: Explicit budgets in evaluation tools do not impose a cloud agent round limit. Other hosts may use their own local storage.

Keep artifacts with their owning host.

Browser-local workspaces and cloud in-memory bundles remain. The hub stores no artifacts.

BROWSER

Local revisions

Keep editable files, revision history, recovery, import/export and portable workspaces.

VIEWERS

Reusable presentation

Retain layouts, previews, controls and reusable artifact viewers.

CLOUD

In-memory bundles

Validate bundles independently and serve sandboxed versions from the cloud artifact actor.

Hardest problem: remove hub storage and publishing across later commits without damaging local artifact workflows.

Merge review: No hub artifact tables, routes, ownership secrets or publishing UI return in this rewrite.

Separate working features from future design.

Soundness checks, authorization proofs and Zig remain independent review decisions.

ASSURANCE

Test the semantics

Generate typed programs, retain regression fixes and compare runtime behavior against an oracle.

AUTHORIZATION

Explain decisions

Keep executable policy examples and proof trees as independently reviewable capabilities.

FUTURE

Cluster questions

Define ownership, replication, partition behavior and effect retry contracts before adding recovery guarantees.

Hardest problem: keep historical event-storming proposals from being mistaken for an accepted disk-first rollout.

Merge review: The study is archival. Its local ledger, snapshot migration and disk vault plan are superseded.

Merge the foundation. Add features in stages.

The final audit covers every retained commit and the dependencies removed with each feature.

FIRST

Cloud foundation

Review in-memory ownership and source checks, then the tested read-only image.

NEXT

Secrets → adapters → language

Keep optional integration and language work after the deployment checkpoint.

THEN

Runtime tools and hosts

Review recording, embedding, local artifacts, assurance and future design independently.

Hardest problem: simplify history without hiding prerequisites or changing the behavior of unrelated later features.

Merge review: Verified: 138 cloud tests, 198 CLI tests and 260 repository EYG tests. Deployment: build, read-only boot and HTTP checks.